Notations

1. Greek Alphabet

Uppercase	Lowercase	Name
A	α	Alpha
B	β	Beta
Γ	γ	Gamma
Δ	δ	Delta
E	ε	Epsilon
Z	ζ	Zeta
H	η	Eta
Θ	θ	Theta
I	ι	Iota
K	κ	Kappa
Λ	λ	Lambda
M	μ	Mu
N	ν	Nu
Ξ	ξ	Xi
O	o	Omicron
Π	π	Pi
P	ρ	Rho
Σ	σ	Sigma
T	τ	Tau
Υ	υ	Upsilon
Φ	φ	Phi
X	χ	Chi
Ψ	ψ	Psi
Ω	ω	Omega

2. Logical & Set Notation

Symbol	Name	Read As	Example
$\neg$	Negation	Not	$\neg P$ (Not P)
$\wedge$	Conjunction	And	$P \wedge Q$ (P and Q)
$\vee$	Disjunction	Or	$P \vee Q$ (P or Q)
$\oplus$	Exclusive Or	Either… or	$P \oplus Q$ (Either P or Q, but not both)
$\rightarrow$	Implication	If… then / Implies	$P \rightarrow Q$ (If P, then Q)
$\leftrightarrow$	Biconditional	If and only if	$P \leftrightarrow Q$ (P if and only if Q)
$\forall$	Universal Quantifier	For all / For every	$\forall x P(x)$ (For all x, $P(x)$ holds)
$\exists$	Existential Quantifier	There exists / For some	$\exists x P(x)$ (There exists an x such that $P(x)$)
$\nexists$	Non-existential Quantifier	There does not exist	$\nexists x P(x)$ (No x satisfies $P(x)$)
$\exists!$	Uniqueness Quantifier	There exists a unique	$\exists! x P(x)$ (There is exactly one x where $P(x)$ holds)
$\top$	Tautology	Top / Always true	$P \vee \neg P \equiv \top$
$\bot$	Contradiction	Bottom / Always false	$P \wedge \neg P \equiv \bot$
$\vdash$	Syntactic Turnstile	Proves / Yields	$\Gamma \vdash P$ (Γ proves P)
$\nvdash$	Non-provability	Does not prove	$\Gamma \nvdash P$ (Γ cannot prove P)
$\vDash$	Semantic Turnstile	Entails / Satisfies	$\Gamma \vDash P$ (Γ semantically entails P)
$\nvDash$	Non-entailment	Does not entail	$\Gamma \nvDash P$ (Γ does not entail P)

Symbol	Name	Read As	Example
$\equiv$	Logical Equivalence	Is logically equivalent to	$P \to Q \equiv \neg P \lor Q$
$\in$	Element of	Belongs to	$x \in A$ (x belongs to set A)
$\notin$	Not Element of	Does not belong to	$x \notin A$ (x is not in set A)
$\subseteq$	Subset	Is a subset of	$A \subseteq B$ (A is contained in or equal to B)
$\subset$	Strict Subset	Is a proper subset of	$A \subset B$ (A is strictly contained in B)
$\not\subset$	Not a Subset	Is not a subset of	$A \not\subset B$ (A is not a subset of B)
$\varnothing$	Empty Set	Empty set / Null set	$A = \varnothing$ (Set A has no elements)
$\cup$	Union	Union of	$A \cup B$ (Elements in A or B or both)
$\cap$	Intersection	Intersection of	$A \cap B$ (Elements in both A and B)
$\setminus$	Set Difference	Minus / Relative complement	$A \setminus B$ (Elements in A but not in B)

Ex.

$$\forall x(x \in A \cup B \to (x \in A \lor x \in B)) \land \nexists y(y \in \varnothing \land y \notin (A \setminus B))$$

Read as: “For all x, x being an element of A union B implies that x is in A or x is in B; and there exists no y such that y is in the empty set and y is not in A minus B.”

Ex.

$$(\Gamma \models P \oplus Q) \equiv (\Gamma \vdash P \vee Q) \wedge (\Gamma \nvdash P \wedge Q) \wedge (P \wedge \neg P \equiv \bot)$$

Read as: “Gamma semantically entails P exclusive-or Q is logically equivalent to saying: Gamma proves P or Q, Gamma does not prove P and Q, and P and not P is a contradiction.”

Ex.

$$(\forall \alpha \in A, \exists! \beta \in B : f(\alpha) = \beta) \rightarrow (A \subseteq B \wedge A \not\subset \varnothing)$$

Read as: “If for every alpha in A there exists a unique beta in B such that f of alpha equals beta, then A is a subset of B and A is not a subset of the empty set.”